

SURVEYING-II

(Max.Marks-100)

PART-A

I. Answer the following questions in one or two sentences each carry 2marks.

1. Define centering of a theodolite.
- 2.Distinguish between latitude and departure of a survey line.
- 3.Define tangential tacheometry.
4. What is a vertical curve.
5. Define aerial photogrammetry. (2x5=10)

PART-B

II. Answer any five questions from the following each carry 6 marks.

1. Explain the temporary adjustment of a transit theodolite.
2. List the fundamental lines of a theodolite.
3. What is meant by omitted measurement. State the different cases.
4. Explain the different rules for balancing the traverse.
5. Explain the theory of stadia tacheometric surveying.
6. State the elements of simple curve.
7. Explain the principles of EDM instruments. (6x5=30)

Answer one full question from the following each carry 15 marks

MODULE-I

- III. a. Describe the measurement of horizontal angle by repetition. 7
- b. Explain the component parts of vernier theodolite. 8

OR

IV. a. Explain the different methods of prolonging a straight line with a theodolite.7

b. List the permanent adjustment vernier theodolite and explain the objectives and its necessities. 8

MODULE-II

V. a. Explain different types of traverse and list the methods of traversing.7

b. The following corrected latitudes and departures correspond to the sides of traverse ABCDE. Compute the independent coordinates. 8

Line	Latitude		Departure	
	Northing	southing	Easting	Westing
AB		364.3	627.00	
BC	440.5		940.74	
CD	991.00			761.90
DE		538.00		777.00
EA		529.2		28.84

OR

VI. a. Explain the procedure for preparing gales traverse table. 7

b. Find the length and bearing of a line AB. The coordinate of points A and B are given below. 8

Points	Coordinate	
	Northing	Easting
A	920.40	785.5
B	1100.00	473.30

MODULE-III

VII. a. Describe the procedure to find out the height of a point whose base is accessible with a theodolite. 7

b. Find out the Reduced level of the top of the tower from the following data. 8

Instrument station.	Reading on BM	Vertical angle.	RL of BM	Distance in m	A&B are in line with
A	1.595	$10^{\circ} 08'$	250.10	30m	Top of
B	1.485				Tower.

OR

VIII. a. Explain the different systems of stadia tacheometry. 7

b. A tacheometer with multiplying constant 100 and additive constant zero was set up at a station O and the following results were obtained by keeping the staff vertical. Calculate the horizontal distance between stations O&P and reduced level of P. 8

Instrument station	Staff station.	Hair readings.	Vertical angle.	Remarks.
O	BM	1.870, 2.200, 2.520.	$+7^{\circ} 00'$	RL of BM
		1.650, 1.810, 1.960.	$-11^{\circ} 00'$	150.75.

MODULE-IV

IX. a. Explain the procedure for setting out simple curve by offset method. 7

b. Two straight AB & BC intersect at a chainage 125 chain +12m. The angle of intersection is 140° . It is required to set out a 5° simple curve to connect the two straight. Calculate all the necessary data to set out the curve by offset from the chord produced with peg interval of 30m. 8

OR

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| X. a. Explain the basic function of EDM instrument. | 5 |
| b. Write the applications of GIS in civil engineering. | 5 |
| c. List the component parts of a total station. | 5 |